QP code: 26900104

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (Honours) REGULAR EXAMINATION October 2026

Fifth SEMESTER

Discipline Specific Elective (DSE) - MG5DSECHE300 -Quantum mechanics, Spectroscopy & Group Theory (2024 ADMISSION ONWARDS)

Duration: 2 hours **Maximum Marks: 70**

INSTRUCTIONS TO THE EXAMINERS

- This scheme is only a guideline for valuation. Any other attempt which is equally valid and correct, may be given due credit.
- Some of the answers given in the scheme are elaborate and this is to help the examiner in their valuation. Please do not expect all those points in the answers of the students.
- ***If the answer is complete full credit may be given even if the required number of valuation points could not be identified***

Part A

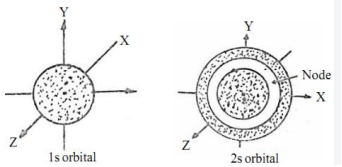
Very Short Answer Questions Answer any 10 questions, each question carries 2 marks

1. Write the time-independent Schrodinger wave equation, in cartesian coordinates [U]/[CO1]

$$-\frac{\hbar^2}{2m}\left(\frac{\partial^2\psi}{\partial x^2}+\frac{\partial^2\psi}{\partial y^2}+\frac{\partial^2\psi}{\partial z^2}\right)+V(x,y,z)\psi=E\psi$$

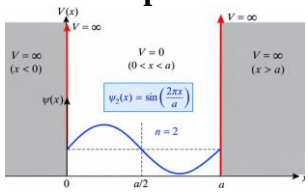
Any correct form with explanation of each term -2 Marks

2. Sketch the shapes of 1s and 2s orbitals. [K]/[CO1]



1 Mark for each diagram

3. Sketch the wavefunction corresponding to quantum number n = 2 for a particle in a one - dimensional box with infinite potential walls.:. [A]/[CO1]



Correct diagram with boundary conditions-2 Marks

4. Write the expression for the number of vibrational degrees of freedom of a nonlinear molecule. [U]/[CO2]

3N-6, Where Nis the number of atoms. ***Equation-2 Marks***

5. Explain the different regions of electromagnetic spectrum used in spectroscopy: [U]/[CO2]

Radio waves- NMR, Microwave- Rotational Spectra, IR- Vibrational Spectra, UV-Visible-Electronic Spectra

½ mark for each region

6. What is meant by a harmonic oscillator in vibrational spectroscopy? [U]/[CO2]

A harmonic oscillator in vibrational spectroscopy is a model in which two atoms bonded together vibrate about their equilibrium positions, behaving like two masses connected by a spring that obeys Hooke's law.

Definition-2 Marks

7. Show how many spin states are possible for Carbon-13 [U] / [CO3]

Number of spin states= 2I+1, I=1/2, 2I+1 = 2 ***Equation- 1 Mark, correct no. of spin states- 1 Mark***

8. What is meant by a singlet electronic state? How does it differ from a triplet state? [K] / [CO3]

Singlet electronic state - total spin quantum number(S=0), - electron spins are paired in opposite directions-Spin multiplicity (2S+1=1). ***- 1 Mark***

A triplet electronic state - total spin quantum number=3), two electrons have parallel spins. Spin multiplicity (2S+1=3).

- 1 Mark

9. List two types of unsaturated functional groups that exhibit magnetic anisotropy. [K] / [CO3]

Carbonyl group (C=O), Carbon–carbon double bond (C=C) in alkenes. ***1 mark for each functional group***

10. Identify the point groups of the molecules Cyclohexane (boat) and Cis-[PtCl2Br2]. [A] / [CO4]

Beyond scope of the syllabus. Give full credit. ***2 marks***

11. Identify the symmetry operations generated by the C3 axis. [A] / [CO4]

C_3^1, C_3^2 ***1 mark for each***

12. Identify the point groups of a centrosymmetric linear molecule and a non-centrosymmetric linear molecule, giving one example of each. [A] / [CO4]

Centrosymmetric linear molecule: Dαh-carbon dioxide).non-centrosymmetric linear molecule $C\alpha V$ −HCl

Point groups ½ Mark each, Examples ½ marks each

[2x10 = 20]

Part B-

Short Answer Questions

Answer any 6 questions, each question carries 5 marks

13. What are spherical polar coordinates? Write the time - independent Schrödinger equation for hydrogen atom in terms of spherical polar coordinates. [U] / [CO1]

Spherical polar coordinates locate a point in three-dimensional space using a radial distance (r) from the origin, a polar angle (Θ) from the positive (z)-axis, and an azimuthal angle φ from the positive (x)-axis to the projection of radial line in the (xy)-plane. x= rsinΘcosφ , y= rsinΘsinφ , z= rcosφ. Also, $r^2 = x^2+y^2+z^2$

$$\frac{1}{r^2}\frac{\partial}{\partial r}\left(r^2\frac{\partial\psi}{\partial r}\right)+\frac{1}{r^2 Sin\theta}\frac{\partial}{\partial\theta}\left(Sin\theta\frac{\partial\psi}{\partial\theta}\right)+\frac{1}{r^2\,Sin^2\theta}\frac{\partial^2\psi}{\partial\phi^2}+\frac{8\pi^2\mu}{h^2}\left(E+\frac{Ze^2}{r}\right)\psi=0$$

(spherical polar coordinates- 3 marks, Schrödinger equation with explanation of terms- 2 Marks)

14. What are quantum mechanical operators. Discuss their significance. [U] / [CO1]

Quantum mechanical operators are mathematical operations that act on wavefunctions to obtain measurable physical quantities, called observables, such as energy, momentum and angular momentum. Quantum mechanical operators are used to represent physical observables such as position, momentum, and energy. They help determine the possible measurable values of physical quantities by operating on wavefunctions. They are essential for calculating expectation values and understanding the uncertainty principle and the behaviour of microscopic particles. Some examples are

Observable	Operator Symbol	Expression
Position	$\hat{\mathbf{x}}$	Multiply by x
Momentum in x-axis	$\hat{\mathbf{P}}_x$	$-i\hbar\frac{\partial}{\partial x}$

(Give full credit for any five relevant points-operators definition, examples and significance) 2 marks

15. Explain how rotational spectroscopy can be applied to determine the bond length of a diatomic molecule. [U] / [CO2]

For a rigid rotor, the separation between adjacent lines in the rotational spectrum is (2B), where (B) is the rotational constant expressed in wavenumbers. The rotational constant is related to the moment of inertia by $B=\frac{h}{8\pi^2 IC}$. But the moment of inertia is related to the reduced mass and bond length by $I=\mu r^2$.Hence bond length $r=\sqrt{\frac{h}{8\pi^2 C\mu B}}$. Where μ is the reduced mass

Expression for rotational constant, internuclear distance & moment of inertia with explanation of terms- 5 Marks

16. What are the limitations of Beer-Lambert's law? [K]/[CO2]

Limitations of Beer-Lambert's law is not mentioned in the syllabus. If Beer-Lambert's law is explained with equations, Give full credit.
5-marks

17. Explain the different types of electronic transitions involving σ, π and n molecular orbitals. Give suitable examples for each transition. [U] / [CO3]

σ → σ* transition: An electron moves from a bonding sigma orbital to an antibonding sigma orbital; it requires the highest energy and occurs in saturated compounds such as ethane (C_2H_6).
n → σ* transition: An electron from a non-bonding orbital moves to an antibonding sigma orbital; it occurs in compounds containing O, N or halogens, such as methanol (CH_3OH).
π → π* transition: An electron moves from a bonding pi orbital to an antibonding pi orbital; it occurs in unsaturated compounds such as ethene (C_2H_4).
n → π* transition: An electron from a non-bonding orbital moves to an antibonding pi orbital; it occurs in compounds containing groups such as carbonyl, for example acetone (CH_3COCH_3).

Different types of electronic transitions with exmaples-5Marks

18. Explain the Laporte selection rule and discuss its significance in electronic spectra of molecules. [U]/[CO3]

The Laporte selection rule states that in centrosymmetric molecules, electronic transitions are allowed only between states of opposite parity, that is, (g → u & u → g).Transitions between states of the same parity, such as (g →g) or (u →u), are forbidden. ***Selection rule -3 Marks***

The rule determines the intensity of electronic absorption bands, with allowed transitions generally being more intense than forbidden transitions. In centrosymmetric transition-metal complexes, (d →d) transitions are Laporte-forbidden because both states have gerade symmetry. These transitions may become weakly allowed through molecular vibrations or distortion that removes the centre of symmetry. **(or any relevant explanation)**

Significance -3 Marks

19. Explain the properties of a mathematical group. [U]/[CO4]

Closure property: The product of any two elements of a group belongs to the same group. Associative property: The group operation satisfies ((AB)C=A(BC)).Identity element: A group contains an identity element (E) such that (AE=EA=A) for every element (A).Inverse element: Every element (A) has an inverse (A^{-1}) such that ($AA^{-1}=A^{-1}A=E$).Commutative property: If (AB=BA) for every pair of elements, the group is called an Abelian group; otherwise, it is a non-Abelian group**.** ***5 Properties- 5 Marks***

20. Using the symmetry operations and group multiplication table of C_{2v}, determine whether the point group is Abelian. Justify your answer [A]/[CO4]

The point group (C_{2v}) contains four symmetry operations: (E), (C_2), (σv) and σv'). ***1 Mark***

C_{2v}	E	$C_{2(z)}$	$\sigma_{v(xz)}$	$\sigma_{v(yz)}$
E	E	$C_{2(z)}$	$\sigma_{v(xz)}$	$\sigma_{v(yz)}$
$C_{2(z)}$	$C_{2(z)}$	E	$\sigma_{v(yz)}$	$\sigma_{v(xz)}$
$\sigma_{v(xz)}$	$\sigma_{v(xz)}$	$\sigma_{v(yz)}$	E	$C_{2(z)}$
$\sigma_{v(yz)}$	$\sigma_{v(yz)}$	$\sigma_{v(xz)}$	$C_{2(z)}$	E

Group Multiplication Table- 2 Marks

A group is Abelian if the product of any two symmetry operations is independent of their order. ***1 Mark***

In C_{2v}, ($C_2 .\sigma v=\sigma v C_2 \sigma v'$). Similarly, all pairs of symmetry operations commute, as confirmed by the group multiplication table. Therefore, (C_{2v}) is an Abelian group ***1 Mark***

Part C

Short Essay

Answer any 2 questions, each question carries 10 marks

21. Elaborate on the black body spectrum and the Planck distribution law. Discuss their significance in development of quantum mechanics. [U]/[CO1]

A black body is an ideal object that absorbs all incident radiation and emits radiation whose spectrum depends only on its temperature. The black body spectrum is a continuous distribution of radiation intensity over different wavelengths, with the intensity maximum shifting towards shorter wavelengths as temperature increases. Classical theories, such as the Rayleigh–Jeans law, failed to explain the observed spectrum at short wavelengths, leading to the ultraviolet catastrophe. Max Planck proposed that energy is emitted or absorbed in discrete packets called quanta, with energy (E=hν). Planck's distribution law is expressed as

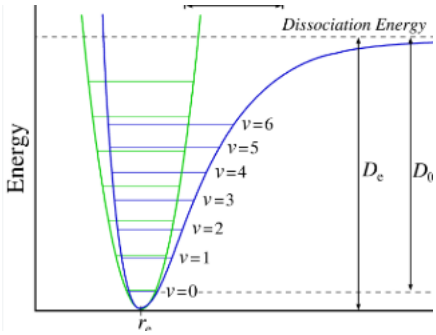
$E_\lambda = \frac{2\pi hc^2}{\lambda^5} \frac{1}{e^{hc/(\lambda kT)} - 1}$ This law successfully explains the observed black body spectrum at all wavelengths and temperatures. Planck's introduction of energy quantization resolved the black body radiation problem and

marked the beginning of quantum mechanics.

(Plank distribution law is not mentioned in the syllabus. Give full credits for relevant points related to blackbody radiation and its role in the development of quantum mechanics.)

(7 marks + 3 marks)

22. Explain the concept of anharmonicity in molecular vibrations and show how the dissociation energy of a diatomic molecule can be determined using the Morse potential. [U]/[CO2]



Anharmonicity refers to the deviation of molecular vibrations from the ideal harmonic oscillator model, in which the restoring force is not perfectly proportional to the displacement from the equilibrium bond length.

- In a harmonic oscillator, the potential energy curve is parabolic, and the vibrational energy levels are equally spaced.
- In a real molecule, the potential energy curve becomes flatter as the bond is stretched, and the energy levels get closer together at higher vibrational quantum numbers. At sufficiently high energy, the bond dissociates, allowing the atoms to separate.

The Morse potential describes the anharmonic potential energy of a diatomic molecule: $V(r) = D_e \left[1 - e^{-a(r-r_e)}\right]^2$ where (D_e) is the dissociation energy measured from the potential minimum, (r_e) is the equilibrium bond length, and (a) is a constant. The vibrational energy levels of an anharmonic oscillator are approximately given by $G(v) = \omega_e \left(v + \frac{1}{2}\right) - \omega_e x_e \left(v + \frac{1}{2}\right)^2$ where (v=0,1,2…..), (ω_e) is the anharmonic vibrational wavenumber, and (x_e is the anharmonicity constant, with the energies expressed in cm^{-1}.

P.E diagram for harmonic and an harmonic oscillator- 2 Marks, Anharmonic Oscillator Definition, Morse function, Expression for Vibrational energy, Dissociation energy with proper Explanation-8 Marks

23. Explain the physical and electronic mechanisms of shielding and deshielding. Detail how these localized phenomena modify the Larmor frequency. [U]/[CO3]

Shielding occurs when electrons surrounding a nucleus circulate in an applied magnetic field and generate a local induced magnetic field that opposes the external field. This reduces the effective magnetic field experienced by the nucleus, making it shielded. Deshielding occurs when the electron density around a nucleus decreases or the induced field reinforces the external field, increasing the effective magnetic field experienced by the nucleus. The effective magnetic field is expressed as $B_{eff}=B_0(1-\sigma)$ where B_0 is the external magnetic field and σ is the shielding constant. The Larmor frequency is given by $\nu = \frac{\gamma B_{eff}}{2\pi}$, where ϒ is the gyromagnetic ratio.

Therefore, shielding decreases the Larmor frequency, whereas deshielding increases it at a constant external magnetic field.

Shielding & Deshielding with any proper explanation- 7 marks, Larmor frequency – 3 Marks

24. Explain matrix representation of symmetry operations. Write down the matrix representation for E, i, Sn and σ. [U]/[CO4]

Symmetry operations can be represented in terms of matrices.

Take symmetry operation 'i'. The inversion operation, i, transforms a coordinate point (x,y,z) into (x',y',z') which is equal to (-x, -y, -z).

x'= (-1) x+0y+0z

y'= 0x+(-1) y+0z

z'= 0x+0y+(-1) z

$\begin{bmatrix} x' \\ y' \\ z' \end{bmatrix} = \begin{bmatrix} -1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & -1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix}$. $\begin{bmatrix} -1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & -1 \end{bmatrix}$ is the matrix representation of i.

2 marks. Give credit if any other symmetry operation is used as example.

Matrix representation of

E= $\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$; i= $\begin{bmatrix} -1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & -1 \end{bmatrix}$; S_n = $\begin{bmatrix} cos\theta & -sin\theta & 0 \\ sin\theta & cos\theta & 0 \\ 0 & 0 & -1 \end{bmatrix}$ or $\begin{bmatrix} cos\theta & sin\theta & 0 \\ -sin\theta & cos\theta & 0 \\ 0 & 0 & -1 \end{bmatrix}$;

σ = $\begin{bmatrix} cos2\beta & sin2\beta & 0 \\ sin2\beta & -cos2\beta & 0 \\ 0 & 0 & 1 \end{bmatrix}$ or σ_{xy} = $\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -1 \end{bmatrix}$ **(any one)**

For each matrix representation – 2 marks each

[10x2 = 20]